Week 8 Level 3: Advanced

Data Detectives

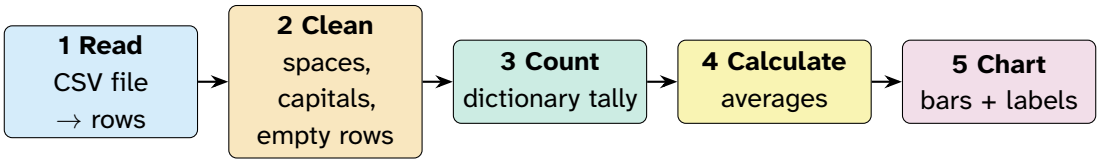
Data tells stories when you clean it, count it and chart it.

What you will learn

- How to read a **CSV file** with Python's `csv` module.
- How to **clean** messy data: extra spaces, MiXeD capitals, empty rows, missing numbers.
- How to **count** with a dictionary and work out an **average**.
- How to draw a **bar chart** that everyone can read – and how to keep survey answers private.

The big idea: the data detective's pipeline

A data detective always works in the same order. Skipping a step gives wrong answers!



1. Read: what is a CSV file?

CSV means *comma-separated values*: a table saved as plain text. The first line is the **header** (the column names). Every other line is one **row**. Here is the start of `favorite_games.csv` – made-up, anonymous survey answers (a response number, no names):

```
response,age_group,favorite_game,genre,hours_per_week,stars
1,10-11,Block Craft,building,5,5
2,12-13, block craft ,building,3,4
5,,,,,
7,12-13,Kart Dash,racing,,3
```

`csv.DictReader` turns each row into a **dictionary**: `row["favorite_game"]` is `"Block Craft"`.

```
import csv
with open("favorite_games.csv", newline="") as f:
```

```
    for row in csv.DictReader(f):
        print(row["favorite_game"], row["stars"])
```

2. Clean: wash the data first

Every value arrives as **text**, even numbers: `"5"`, not 5. And people type things in different ways. Spot the mess:

Messy value	Problem	Clean it with
`" block craft "`	extra spaces, small letters	`.strip().title()` → `"Block Craft"`
`"BLOCK CRAFT"`	capitals	`.title()` → `"Block Craft"`
`"5"`	a number stored as text	`int("5")` → 5
`""` (hours)	missing answer	use `None`, and skip it in averages
`5,,,,,`	empty row	skip the whole row

3 and 4. Count and calculate

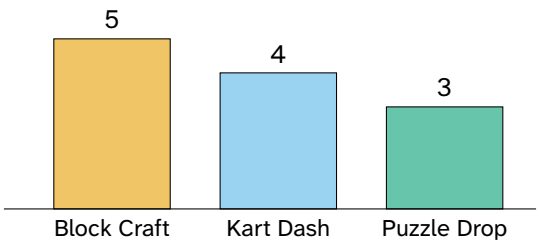
Counting with a dictionary is like a tally chart. The first time you see a game, `counts.get(game, 0)` gives `0`:

```
counts = {}
for row in rows:
    game = row["favorite_game"]
    counts[game] = counts.get(game, 0) + 1
# {"Block Craft": 5, "Kart Dash": 4, ...}
```

The **average** (mean) is the total divided by how many: `sum(numbers) / len(numbers)`. Leave out missing answers – counting a blank as `0` would make the average unfairly small.

5. Chart: readable by everyone

A good chart never relies on color alone. Write the **name** under each bar and the **number** on top, so someone who sees colors differently can still read it.



Exercise 8.1: Survey cleaner

Folder: `week08/ex-survey-cleaner`. Open `favorite_games.csv` in Thonny and circle the mess you find. Then finish `to_number()`, `is_empty()` and add `.title()` to the game names. **Done:** `Clean answers: 17`.

How many rows did the cleaner throw away, and why?

Exercise 8.2: Survey stats

Folder: `week08/ex-survey-stats`. Finish `count_by()`, `average()` and `most_popular()`. Then answer from your program's report:

Most popular game: _____ Average hours per week: _____

Adventure games got 5.0 stars on average from only 2 answers. Should we trust that? Why?

Exercise 8.3: Bar chart

Folder: `week08/ex-bar-chart`. Make `bar_height()` scale the bars so the biggest is `MAX_HEIGHT` pixels, and write a number on top and a name under every bar. **Done:** six labeled bars from Block Craft (5) to Space Chess (1).

Digital citizenship & safety: Consent and privacy in surveys

Before collecting data, ask: *Did people agree to answer? Can they skip a question? Do I really need this question?* Never collect real names, addresses, schools, birthdays or photos. Even without names, answers together can point to one person ("the only 13-year-old in the chess club"). Before sharing data, **anonymize** it: keep only the columns you need and give each row an ID like R1, R2.

Vocabulary box

CSV Comma-separated values: a table saved as plain text, one row per line.

header The first line of a CSV file, with the column names.

data cleaning Fixing messy data (spaces, capitals, missing or empty values) before using it.

average (mean) The total divided by how many values there are.

sample The group of people who answered. A small sample can give misleading results.

consent Agreeing, freely, to take part – with the choice to say no.

anonymize Remove or replace anything that could show who gave an answer.

Reflection

What story does the bar chart tell? What can it *not* tell us?

Would you answer a survey that asked for your full name? Why or why not?

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15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Anonymize a survey (about 10 minutes)

Open `3-advanced/exercises/week08/hw-survey-privacy/starter.py` and press F5. The file `snack_survey.csv` is made up, but it has a `nickname` and a `club` column. Together they could show exactly who answered.

1. Finish `anonymize(rows, keep)`: each new row gets an `id` (R1, R2, ...) and only the columns in `keep`.
2. Run again. The first row should be `{'id': 'R1', 'favorite_snack': 'apple slices', 'minutes_reading': '20'}`.
3. Run `check_my_work.py`.

Why is `club` risky even without a name?

_____

★★ Challenge (want more?): Save the safe file

Finish `save_csv(rows, filename)` with `csv.DictWriter`: open the file with `"w"` and `newline=""`, write the header, then the rows. Run the program and open `snack_survey_anonymous.csv` in Thonny (**File** → **Open**, choose "All files"). Which snack is the most popular?

_____

Creative (make it yours): Design a kind, safe survey

Plan a 4-question survey for your family about a fun topic (favorite board game, best pizza topping, dream pet). Rules: start with a consent sentence ("You can skip any question"); no names, ages, addresses or schools; at least one question with a number answer you could average.

Self-check: tick each box when it is true

- ☐ I know that CSV values arrive as text and must be converted to numbers.
- ☐ I can explain why we skip empty answers instead of counting them as `0`.
- ☐ My anonymous rows have IDs and no nickname or club.
- ☐ I know to ask for consent and collect only what I need.